SchemeSathi: An AI-Powered RAG-Based Assistant for Government Scheme Discovery and Simplification

Problem Statement

Information about government welfare schemes in India is scattered across numerous central and state department websites, portals, notifications and PDF circulars, each using inconsistent formats, structures and administrative language. This makes it difficult for an ordinary citizen to discover the schemes they are eligible for, or to correctly interpret eligibility conditions, benefits, required documents and application procedures. This project builds SchemeSathi, an AI-powered scheme assistant that periodically crawls a curated set of official sources, detects newly published or modified information, and processes the resulting webpages and PDFs into a continuously updated Retrieval-Augmented Generation (RAG) knowledge base. Given a user's basic profile (state, occupation and category) and a natural-language query, the system retrieves the most relevant scheme content from this knowledge base and uses a large language model (LLM) to generate a simplified explanation of eligibility, benefits, documents and application steps, accompanied by a link to the official source so that the answer can be independently verified.

Abstract

A large number of citizens who are eligible for government welfare schemes never apply for them, primarily because information about these schemes is fragmented across many official websites and documents, written in dense bureaucratic language, and updated without a central point of announcement. Existing scheme-discovery portals typically rely on manual curation or simple keyword search, which quickly becomes outdated and cannot answer specific, natural-language questions asked by individual users.

SchemeSathi addresses this problem with an automated pipeline that combines web crawling, change detection, document processing and Retrieval-Augmented Generation. A scheduled crawler collects webpages and PDF notifications from a curated list of official government sources; a change-detection module compares each fetched document against its previously stored version to identify new or updated schemes without re-processing unchanged content. Updated documents are cleaned, parsed and split into semantically coherent chunks, converted into vector embeddings, and stored in a vector database that forms the system's continuously refreshed knowledge base. When a user supplies their state, occupation and category and asks a question in plain language, the system performs semantic retrieval over this knowledge base, filters results by the user's profile, and passes the retrieved context to an LLM, which composes a simplified, structured answer covering eligibility, benefits, required documents and the application procedure, together with a citation to the originating official source.

By combining automated freshness, profile-aware retrieval and LLM-based simplification with source citation, SchemeSathi aims to make government scheme discovery faster, more accurate and independently verifiable for citizens who would otherwise struggle to navigate scattered official information.

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